

# Food Delivery Analytics

## Overview

An end-to-end data analytics project examining **15,000 food delivery orders** across **36 variables** to uncover operational inefficiencies, customer behaviour patterns, and revenue opportunities.

The analysis was completed entirely in **Google Sheets** using formulas, lookup tables, dashboards, and pivot tables—demonstrating how meaningful business insights can be generated without programming tools.

## Business Questions

- What factors have the greatest impact on delivery performance?
  - Which customer segments generate the most revenue?
  - How do traffic and weather affect operational efficiency?
  - Are loyalty and premium programmes driving meaningful behavioural change?
  - Which areas should the business prioritise to improve customer satisfaction and profitability?
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## Project Structure

food-delivery-analytics/

```
|  
|— README.md  
|— INSIGHTS.md  
  
|  
|— data/  
|   |— food_delivery_analytics_cleaned.csv  
  
|— analysis/  
|   |— food_delivery_analysis_v2.xlsx
```

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## Dataset Overview

| Property            | Detail                  |
|---------------------|-------------------------|
| Dataset             | Food Delivery Orders    |
| Records             | 15,000                  |
| Variables           | 36                      |
| City Tiers          | Metro, Urban, Sub-urban |
| Time Period         | 12 Months               |
| Revenue<br>Analysed | \$1.79M                 |
| Tooling             | Google Sheets           |

## Key Variables

### Customer

- Age
- Loyalty Score
- Loyalty Tier
- Premium Customer Flag

### Financial

- Order Value
- Delivery Fee
- Discount Amount
- Tip Amount
- Final Amount Paid

### Delivery

- Delivery Time
- Delivery Distance
- Preparation Time
- Estimated Delivery Time

### Operations

- Traffic Severity
- Weather Severity
- Delivery Efficiency Score
- Partner Experience

## Flags

- Cancellation
  - Delay
  - Refund
  - Promo Code Usage
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## Key Findings

- Generated **\$1.79M** revenue from **15,000 orders**
- Average order value: **\$113.95**
- Cancellation rate: **13.35%**
- Tier 3 cities generated **50.2% of total revenue**
- Premium customers spent **12.2% more per order**
- High traffic increased delivery times by approximately **13 minutes**
- Severe weather increased delivery times by approximately **11 minutes**
- Loyalty tier showed minimal impact on customer behaviour
- Late-night orders experienced the highest cancellation rates

See **INSIGHTS.md** for the complete analysis and business recommendations.

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## Tools & Techniques

### Tools

- Google Sheets

### Functions

- COUNTIF / COUNTIFS
- SUMIF / SUMIFS
- AVERAGEIF / AVERAGEIFS
- IFERROR
- XLOOKUP
- IFS
- Pivot Tables

## **Analytical Methods**

- KPI Analysis
  - Customer Segmentation
  - Cohort Comparison
  - Correlation Analysis
  - Cross-tabulation
  - Trend Analysis
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## **Methodology**

1. Reviewed and validated all source variables.
  2. Created lookup tables for customer, traffic, weather, and experience bands.
  3. Built KPI dashboards using formula-driven calculations.
  4. Performed segmentation analysis across customer and operational groups.
  5. Created pivot tables to identify trends and relationships.
  6. Conducted correlation analysis to identify key delivery drivers.
  7. Converted findings into prioritised business recommendations.
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## **Deliverables**

- Interactive Excel/google Workbook
- KPI Dashboard
- Pivot Table Analysis
- Business Insights Report
- Strategic Recommendations